

Verifying Forces sum to 0

Pin C

$$\Sigma F_x = CD + CE \frac{a/2}{h} - BC \frac{a}{d} = 0$$

$$= 0 + (-20.031) \frac{0.4/2}{0.361} - (-13.889) \left(\frac{0.4}{0.5} \right) = 0$$

$$0 = 0 \checkmark$$

$$\Sigma F_y = \cancel{BC} \frac{b}{d} + CE \cdot \frac{b}{h} + P = 0$$

$$= (-13.889) \left(\frac{0.3}{0.5} \right) + (-20.031) \left(\frac{0.3}{0.361} \right) + 25 = 0$$

$$0 = 0 \checkmark$$

Pin D

$$\Sigma F_x = AD \frac{a}{d} - CD - DE \frac{a/2}{h} = 0$$

$$\Rightarrow (13.889) \left(\frac{0.4}{0.5} \right) - 0 - (20.031) \left(\frac{0.4/2}{0.361} \right) = 0$$

$$0 = 0 \checkmark$$

$$\Sigma F_y = DE \frac{b}{h} + AD \frac{b}{d} - P = 0$$

$$\Rightarrow (20.031) \left(\frac{0.3}{0.361} \right) + (13.889) \left(\frac{0.3}{0.5} \right) - 25 = 0$$

$$0 = 0 \checkmark$$

Pin A

$$\Sigma F_x = AE + AD \frac{a}{d} = 0$$

$$\Rightarrow \cancel{13.889} + (-11.111) + (13.889) \left(\frac{0.4}{0.5} \right) = 0$$
$$0 = 0 \checkmark$$

$$\Sigma F_y = Ay - AD \cdot \frac{b}{d} = 0$$

$$\Rightarrow 8.333 - (13.889) \cdot \left(\frac{0.4}{0.5} \right) = 0$$
$$\Rightarrow 0 = 0 \checkmark$$

Pin B

$$\Sigma F_x = BE + BC \frac{a}{d} = 0$$

$$\Rightarrow \cancel{11.111} + (-13.889) \left(\frac{0.4}{0.5} \right) = 0$$
$$0 = 0 \checkmark$$

$$\Sigma F_y = -By - BC \frac{b}{d} = 0$$

$$\Rightarrow -(8.333) - (-13.889) \left(\frac{0.3}{0.5} \right) = 0$$

$$0 = 0 \checkmark$$